1. Find the sinusoids corresponding to these phasors:
   (a) V = -25∠40° V
   (b) I = j(12-j5)   A
   (c) I = -3+4j   A
   (d) $V = j8e^{-j20°}$ V

2. Calculate the phase angle between $v_1 = -10\,cos(\omega t + 50°)$ and $v_2 = 12\,sin(\omega t - 10°)$. State which sinusoid is leading.

3. Evaluate the following complex number and write the expression in rectangular form and phasor form both
   (a) $(40\angle 50° + 20\angle -30°)^{\frac{1}{2}}$
   (b) $\frac{10\angle -30° + (3-j4)}{(2+j4)(3-j5)^*}$ (Where * represents conjugate)
   (c) $[(5+2j)(1+4j) - 5\angle 60°]^*$ (Where * represents conjugate)
   (d) $\frac{10+5j+3\angle 40°}{-3+j4} + 10\angle 30° + j5$

4. Determine the input impedance of the circuit in Fig. 1 at ω = 10 rad/s.

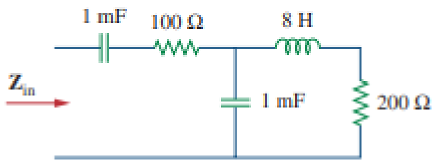

Fig. 1

5. Calculate v_0 in the circuit of Fig. 2.

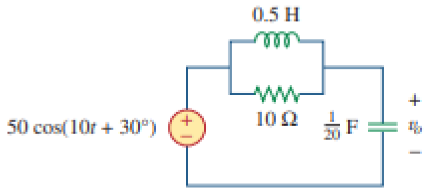

Fig. 2

6. Find v_1 and v_2 in the given circuit as in Fig. 3.

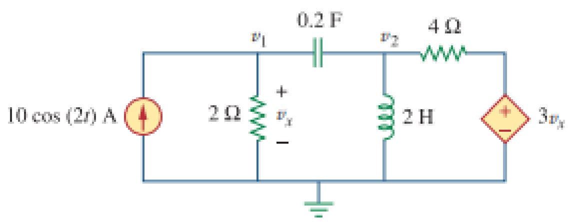

Fig. 3

7. Find the Thevenin equivalent across the terminal a-b as shown in Fig. 4.

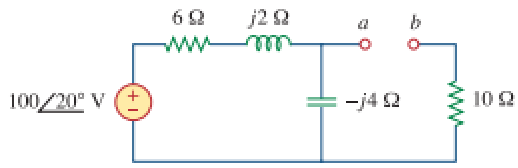

Fig. 4

8. Obtain current I_0 in Fig. 5 using Norton theorem.

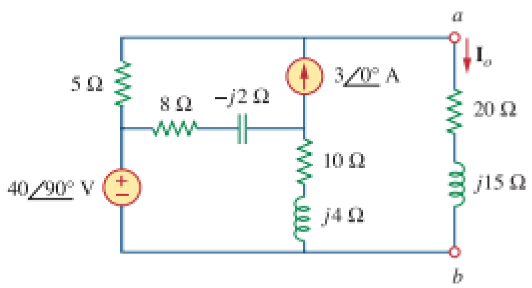

Fig. 5

9. Calculate the instantaneous power and average power for the given voltage and current expressions:

$$v(t) = 330\, cos(10t + 20°)\ \ V$$
$$i(t) = 33\, sin(10t + 60°)\ \ A$$

10. For the circuit shown in Fig. 6, find the load impedance that absorbs the maximum active power. Calculate that maximum active power.

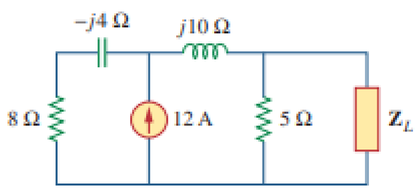

Fig. 6

11. Obtain the power factor and the apparent power of a load whose impedance is $Z = 60 + j40\ \Omega$ when the applied voltage is $v(t) = 320\, cos(377t + 10°)$ V.

12. For a load, $V_{rms} = 110\angle 85°\ \ V$, $I_{rms} = 0.4\angle 15°\ \ A$. Determine: (a) the complex and apparent powers, (b) the real and reactive powers, and (c) the power factor and the load impedance.

13. Find the value of parallel capacitance needed to correct a load of 140 kVAR at 0.85 lagging pf to unity pf. Assume that the load is supplied by a 110 V (rms), 60-Hz line.